I checked your uploaded Python file, Nested Loop Examination Seating Generator.py. I'll prepare the complete activity record based on your actual code, using simple language suitable for a B.Tech 1st-year CSE Data Science student.

Your program generates an examination seating arrangement using nested loops. It displays seat numbers according to the number of rows and seats per row, and marks specified reserved seats with R.

Nested Loop Examination Seating Generator

1. Problem statement

To develop a Python program that generates an examination seating arrangement based on the number of rows and seats per row. The program should display each seat number and identify reserved seats using the letter R.

2. Objectives

1. To generate an examination seating arrangement.
2. To use nested for loops to display rows and seats.
3. To accept reserved seat numbers from the user.
4. To identify reserved seats and display them as R.
5. To understand lists, strings, loops, and conditional statements in Python.

3. Assumptions

- The user enters the number of rows and seats per row as integers.
- Rows and columns are numbered starting from 1.
- Seat numbers are represented as row-column, such as 1-2.
- Multiple reserved seats are separated by commas.
- If the user enters none or leaves the reservation input blank, no seats are reserved.
- Reserved seat numbers must match the format generated by the program.

4. Algorithm

1. Start the program.
2. Input the number of rows.
3. Input the number of seats per row.
4. Input the reserved seat numbers.
5. Create an empty list to store reserved seats.
6. If reserved seats are provided, split them by commas and store them in the list.
7. Display the legend R = Reserved seat.
8. Use an outer for loop to process each row.
9. Use an inner for loop to process each seat in that row.

10. Generate the seat number in row-column format.
11. If the seat is reserved, display R; otherwise, display the seat number.
12. Move to the next line after displaying each row.
13. Stop the program.

5. Flowchart

Start

Input rows, seats per row, and reserved seats

Store reserved seats in a list

Process each row using the outer loop

Process each seat using the inner loop

Is the seat reserved?

Yes

Display R

No

Display seat number

Continue loops until all seats are displayed

End

Note: The reserved-seat check happens for every seat inside the inner loop. After each row, the program starts a new line.

6. Pseudocode

```
BEGIN
   INPUT rows
   INPUT cols
   INPUT reserved

   reserved_seats = empty list

   IF reserved is not "none" AND not empty
      FOR each item in reserved split by comma
         ADD trimmed item to reserved_seats
      END FOR
   END IF
```

```
    DISPLAY "R = Reserved seat"

    FOR r = 1 TO rows
        FOR c = 1 TO cols
            seat = string(r) + "-" + string(c)

            IF seat is in reserved_seats
                DISPLAY "R" without new line
            ELSE
                DISPLAY seat without new line
            END IF
        END FOR

        MOVE to next line
    END FOR
END
```

7. Python source code

```
rows = int(input("Number of rows: "))
cols = int(input("Seats per row: "))
reserved = input("Reserved seats (example 1-2,2-3), or none: ")

reserved_seats = []

if reserved.lower() != "none" and reserved.strip() != "":
    for item in reserved.split(","):
        reserved_seats.append(item.strip())

print("\nR = Reserved seat")

for r in range(1, rows + 1):
    for c in range(1, cols + 1):
```

```
        seat = str(r) + "-" + str(c)

        if seat in reserved_seats:
            print("R", end="\t")
        else:
            print(seat, end="\t")

    print()
```

8. Test cases

Test	Rows	Seats per row	Reserved seats	Expected result
1	3	4	1-2,2-3	Two seats displayed as R
2	2	3	none	All seat numbers displayed
3	1	4	1-1	First seat displayed as R
4	2	2	Blank input	All seats displayed
5	3	3	2-2	Middle seat of the second row displayed as R

9. Sample output screenshots

These are sample console output previews, not actual screenshots captured from a running Python interpreter.

Test 1 — Seating arrangement with reserved seats

```
Number of rows: 3
Seats per row: 4
Reserved seats (example 1-2,2-3), or none: 1-2,2-3

R = Reserved seat
1-1 R 1-3 1-4
2-1 2-2 R 2-4
3-1 3-2 3-3 3-4
```

Copy sample output

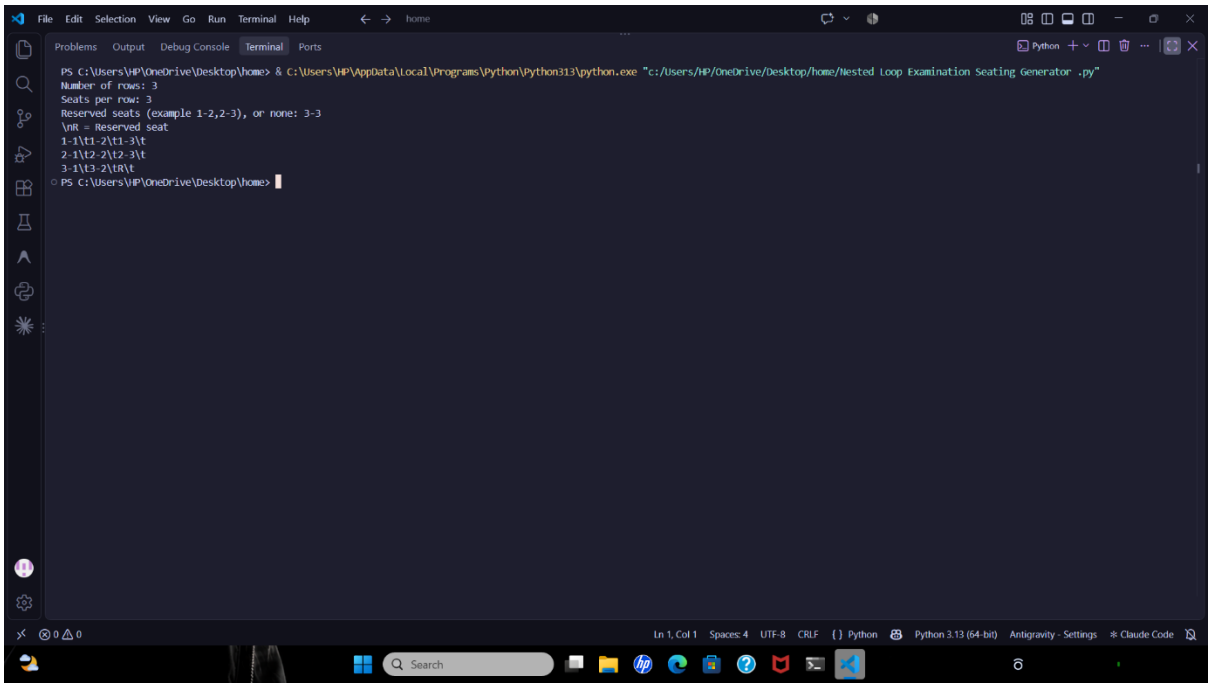
Test 2 — No reserved seats

```
Number of rows: 2
Seats per row: 3
Reserved seats (example 1-2,2-3), or none: none

R = Reserved seat
1-1 1-2 1-3
2-1 2-2 2-3
```

Copy sample output

To obtain actual screenshots for your activity record, run the program in Python IDLE or VS Code and capture the output window.



10. Conclusion

The Nested Loop Examination Seating Generator was successfully developed using Python. It generates seat numbers based on the given number of rows and seats per row. It identifies reserved seats and displays them using the letter R. This program helps in understanding nested loops, lists, string operations, conditional statements, and formatted output in Python.